Backward Difference Method for Parabolic Equations

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1 Parabolic Partial Differential Equations

The parabolic partial differential equation, or heat equation, also known as the diffusion equation, is

$$\frac{\partial u}{\partial t}(x,t) = \alpha^2 \frac{\partial^2 u}{\partial x^2}(x,t), \qquad 0 < x < l, \qquad t > 0. \tag{1}$$

It is subject to the conditions

$$u(0,t) = u(l,t) = 0, \qquad t > 0, \qquad \text{and} \qquad u(x,0) = f(x). \tag{2}$$

The approach used to solve Eq. **??** is the same as the one presented in [**?**] and [**?**]. A grid is defined by selecting an integer $m > 0$ and setting the spatial step size to $h = l/m$. A temporal step size k is then selected. The grid points are (x_i, t_j), where $x_i = ih$ for $i = 0, 1, \ldots, m$ and $t_j = jk$ for $j = 1, 2, \ldots$.

1.1 Backward Difference Method

When the forward-difference method is used to solve parabolic partial differential equations, unstable systems may be obtained; see [**?**]. To obtain an unconditionally stable method, an implicit finite-difference approximation is considered. The backward approximation for the time derivative is

$$\frac{\partial u}{\partial t}(x_i, t_j) = \frac{u(x_i, t_j) - u(x_i, t_{j-1})}{k} + \frac{k}{2}\frac{\partial^2 u}{\partial t^2}(x_i, \mu_j), \tag{3}$$

where $\mu_j \in (t_{j-1}, t_j)$.

A finite-difference expression for the second derivative with respect to x is also required. A Taylor expansion gives

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_j) = \frac{u(x_i + h, t_j) - 2u(x_i, t_j) + u(x_i - h, t_j)}{h^2} - \frac{h^2}{12}\frac{\partial^4 u}{\partial x^4}(\xi_i, t_j), \tag{4}$$

where $\xi_i \in (x_{i-1}, x_{i+1})$.

Substituting Eqs. **??** and **??** into Eq. **??** gives

$$\begin{aligned} \frac{u(x_i, t_j) - u(x_i, t_{j-1})}{k} &- \alpha^2 \frac{u(x_{i+1}, t_j) - 2u(x_i, t_j) + u(x_{i-1}, t_j)}{h^2} \\ &= -\frac{k}{2}\frac{\partial^2 u}{\partial t^2}(x_i, \mu_j) - \alpha^2 \frac{h^2}{12}\frac{\partial^4 u}{\partial x^4}(\xi_i, t_j). \end{aligned} \tag{5}$$

The resulting backward-difference method is

$$\frac{w_{i,j} - w_{i,j-1}}{k} - \alpha^2 \frac{w_{i+1,j} - 2w_{i,j} + w_{i-1,j}}{h^2} = 0, \tag{6}$$

for each $i = 1, 2, \ldots, m-1$ and $j = 1, 2, \ldots$. Defining

$$\lambda = \frac{\alpha^2 k}{h^2}, \tag{7}$$

Eq. **??** can be rewritten as

$$(1 + 2\lambda) w_{i,j} - \lambda w_{i+1,j} - \lambda w_{i-1,j} = w_{i,j-1}, \tag{8}$$

for each $i = 1, 2, \ldots, m-1$ and $j = 1, 2, \ldots$.

From the initial and boundary conditions,

$$w_{i,0} = f(x_i), \qquad i = 1, 2, \ldots, m-1, \tag{9}$$

and

$$w_{0,j} = w_{m,j} = 0, \qquad j = 1, 2, \ldots. \tag{10}$$

Therefore, the backward-difference method can be represented in matrix form as

$$\begin{bmatrix} 1+2\lambda & -\lambda & 0 & \cdots & 0 \\ -\lambda & \ddots & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\lambda \\ 0 & \cdots & 0 & -\lambda & 1+2\lambda \end{bmatrix} \begin{bmatrix} w_{1,j} \\ w_{2,j} \\ w_{3,j} \\ \vdots \\ w_{m-1,j} \end{bmatrix} = \begin{bmatrix} w_{1,j-1} \\ w_{2,j-1} \\ w_{3,j-1} \\ \vdots \\ w_{m-1,j-1} \end{bmatrix}. \tag{11}$$

Equivalently,

$$A\mathbf{w}^{(j)} = \mathbf{w}^{(j-1)}, \tag{12}$$

for each time level $j = 1, 2, \ldots$.

This formulation defines an iterative method for approximating the solution as time advances. Because $\lambda > 0$, the matrix A is symmetric, positive definite, strictly diagonally dominant, and tridiagonal. Therefore, Crout factorization is suitable for solving the linear system. A generalization of this method for block-tridiagonal matrices can be found in [**?**].

2 Backward-Difference Algorithm for the Heat Equation

The development in the previous section provides the tools required to construct an algorithm for solving parabolic partial differential equations of the form given in Eq. **??**. The procedure is as follows:

- Construct the initial linear system in Eq. **??**.
- Solve the system to obtain the next temporal state $\mathbf{w}^{(j)}$ from $\mathbf{w}^{(j-1)}$.
- Repeat the process for the required number of time steps.

2.1 Algorithm for the Parabolic Heat Equation

The backward-difference method can be used to construct an algorithm that approximates the solution of the boundary-value problem defined in Eq. **??**. Algorithm **??** uses the generalized Crout factorization function implemented in [?]. The Python implementation is available in the following repository: `https://github.com/Julio-Medina/Finite_Difference_Method_Parabolic/tree/main/code`.

Algorithm 1 Backward-Difference Method for the Heat Equation

> **function** BACKWARD_DIFFERENCE_METHOD(l, T, α, N, m, f)
> $h \leftarrow l/m$
> $k \leftarrow T/N$
> $\lambda \leftarrow \alpha^2 k/h^2$
> $x \leftarrow$ numpy.linspace$(0, l, m+1)$
> $t \leftarrow$ numpy.linspace$(0, T, N+1)$
> $A \leftarrow$ zeros$(m-1, m-1)$
> $w \leftarrow f(x[1:-1])$
>   **for** $i \leftarrow 0$ to $m-2$ **do**
> $A[i,i] \leftarrow 1 + 2\lambda$
>   **if** $i < m-2$ **then**
> $A[i, i+1] \leftarrow -\lambda$
> $A[i+1, i] \leftarrow -\lambda$
>   **end if**
>   **end for**
>   **for** $j \leftarrow 1$ to N **do**
> $w \leftarrow$ CROUT_GENERALIZATION$(A, w, m-1)$
>   **end for**
>   **return** A, w, x
> **end function**

2.2 Example

Use the backward-difference method with $h = 0.1$ and $k = 0.01$ to approximate the solution of the heat equation

$$\frac{\partial u}{\partial t}(x,t) - \frac{\partial^2 u}{\partial x^2}(x,t) = 0, \qquad 0 < x < 1, \qquad t > 0, \tag{13}$$

subject to the boundary and initial conditions

$$u(0,t) = u(1,t) = 0, \qquad t > 0, \qquad u(x,0) = \sin(\pi x), \qquad 0 \le x \le 1,$$

and compare the approximations with the analytical solution

$$u(x,t) = e^{-\pi^2 t} \sin(\pi x). \tag{14}$$

The pointwise errors are shown in the following table:

i	x_i	$w_{i,50}$	$u(x_i, 0.5)$	$\lvert u(x_i, 0.5) - w_{i,50}\rvert$
0	0.0	0.000000	0.000000e+00	0.000000e+00
1	0.1	0.002898	2.222414e-03	6.756025e-04
2	0.2	0.005512	4.227283e-03	1.285072e-03
3	0.3	0.007587	5.818356e-03	1.768750e-03
4	0.4	0.008919	6.839888e-03	2.079291e-03
5	0.5	0.009378	7.191883e-03	2.186296e-03
6	0.6	0.008919	6.839888e-03	2.079291e-03
7	0.7	0.007587	5.818356e-03	1.768750e-03
8	0.8	0.005512	4.227283e-03	1.285072e-03
9	0.9	0.002898	2.222414e-03	6.756025e-04
10	1.0	0.000000	8.807517e-19	8.807517e-19

3 Conclusions

When the finite-difference method is applied to parabolic partial differential equations, an explicit forward-difference discretization may exhibit unstable behavior unless a stability restriction is satisfied. The implicit backward-difference approximation provides an unconditionally stable alternative. The associated linear system in Eq. **??** has a symmetric, positive-definite, strictly diagonally dominant, tridiagonal coefficient matrix for $\lambda > 0$. Consequently, Crout factorization is an appropriate method for solving the system. The truncation error is of order

$$O(k + h^2). \tag{15}$$

References

[1] Richard L. Burden and J. Douglas Faires, *Numerical Analysis*, 9th edition, Brooks/Cole, Cengage Learning. ISBN 978-0-538-73351-9.

[2] Julio Medina, *Finite Difference Method for Elliptic Equations.* `https://github.com/Julio-Medina/Finite_Difference_Method`

[3] Richard S. Varga, *Matrix Iterative Analysis*, 2nd edition, Springer. DOI: 10.1007/978-3-642-05156-2.